

HDOC - Day 8

(C Programming)

C Programming Activity Report

Question 1 - Cone: Slant Height and Curved Surface Area

1. Problem Understanding, Input, Output and Formulae

Find the slant height and curved surface area of a cone when radius is given in centimeters and height is given in meters. Since the final answer is required in meters, convert radius from centimeters to meters before calculation.

Inputs: radius in centimeters (cm), height in meters (m).

Outputs: slant height in meters (m), curved surface area in square meters (m²).

Formulae:

Radius in meters = radius in cm / 100

Slant Height = $\sqrt{(\text{radius}^2 + \text{height}^2)}$

Curved Surface Area = $\text{PI} \times \text{radius} \times \text{slantHeight}$

Use PI = 3.14.

2. Standard Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	radius=30 cm, height=0.4 m	Slant=0.50 m; CSA=0.47 m ²	0.50 m; 0.47 m ²	PASS
2	radius=60 cm, height=0.8 m	Slant=1.00 m; CSA=1.88 m ²	1.00 m; 1.88 m ²	PASS
3	radius=50 cm, height=1.2 m	Slant=1.30 m; CSA=2.04 m ²	1.30 m; 2.04 m ²	PASS
4	radius=0 cm, height=2 m	Slant=2.00 m; CSA=0.00 m ²	2.00 m; 0.00 m ²	PASS

3. Algorithm and Evaluation

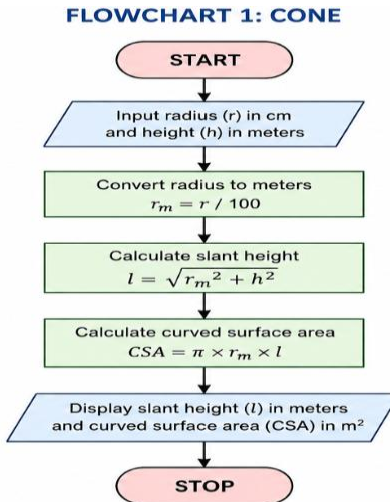
1. Start.
2. Declare radius, height, slantHeight and curvedSurfaceArea.
3. Define PI = 3.14.
4. Input radius in centimeters and height in meters.
5. Convert radius to meters: radius = radius / 100.
6. Calculate slantHeight = $\sqrt{\text{radius} \times \text{radius} + \text{height} \times \text{height}}$.
7. Calculate curvedSurfaceArea = $\text{PI} \times \text{radius} \times \text{slantHeight}$.
8. Display slantHeight and curvedSurfaceArea.

9. Stop.

Evaluation: The algorithm first converts the radius to meters so both dimensions use the same unit. The calculated outputs match the prepared test cases.

4. Flowchart and Evaluation

Flowchart logic: Start → Input radius (cm) and height (m) → Convert radius to meters → Calculate slant height → Calculate curved surface area → Display results → Stop.



Evaluation: The flowchart follows the same sequence as the algorithm and produces the expected results for the test cases.

5. C Program

```
#include <stdio.h>
#include <math.h>

#define PI 3.14

int main()
{
    float radius, height;
    float slantHeight, curvedSurfaceArea;

    printf("Enter radius in cm: ");
    scanf("%f", &radius);

    printf("Enter height in meters: ");
    scanf("%f", &height);

    radius = radius / 100;

    slantHeight = sqrt(radius * radius + height * height);
    curvedSurfaceArea = PI * radius * slantHeight;

    printf("Slant Height = %.2f m\n", slantHeight);
    printf("Curved Surface Area = %.2f sq. m\n", curvedSurfaceArea);

    return 0;
}
```

6. Compilation, Execution and Test Evaluation

Compilation Command:

```
cc cone.c -o cone -lm
```

Execution Command:

```
./cone
```

The program output should be compared with the expected values in the test case table. Matching values are marked PASS.

7. Compilation and Execution

```
(kali@kali)-[~/hdoc8]
$ gcc cone.c -o cone -lm
```

```
(kali@kali)-[~/hdoc8]
$ ./cone
Enter radius in cm:30
Enter height in meters:0.4
SlantHeight=0.50m
CSA=0.47sq.m
```

Question 2 - Total Area of a Rectangle with Two Semicircular Ends

1. Problem Understanding, Input, Output and Formulae

Find the total area of a rectangle with two semicircular ends. For this report, length is taken as the length of the rectangular portion and width is taken as the width of the rectangle and the diameter of each semicircular end.

Assumption: The length and width of the rectangular portion are given, and the width of the rectangle is equal to the diameter of each semicircular end.

Inputs: length of rectangular portion, width of rectangle (diameter of semicircles).

Output: total area of the complete shape.

Formulae:

Radius = width / 2

Rectangle Area = length × width

Area of two semicircles = $\text{PI} \times \text{radius} \times \text{radius}$

Total Area = Rectangle Area + Area of two semicircles

Use PI = 3.14.

2. Standard Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result (Pass/Fail)
1	length=10, width=4	Total Area=52.56	52.56	PASS
2	length=20, width=10	Total Area=278.50	278.50	PASS
3	length=8, width=6	Total Area=76.26	76.26	PASS
4	length=15, width=8	Total Area=170.24	170.24	PASS

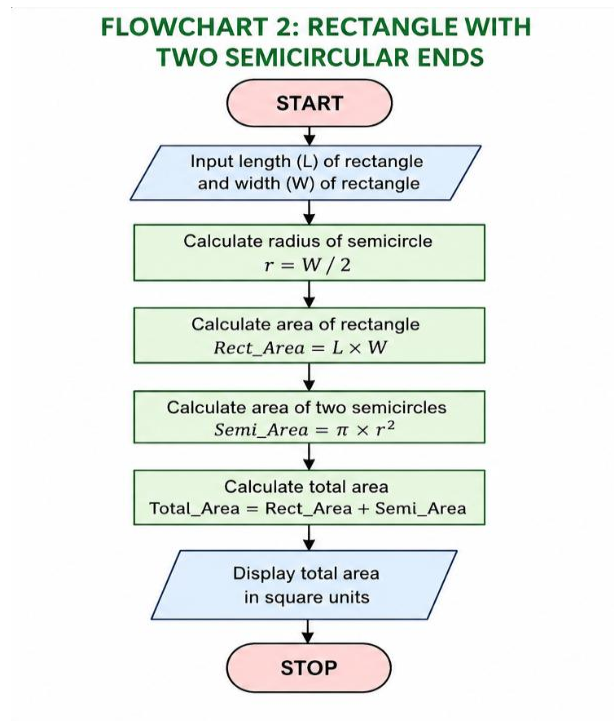
3. Algorithm and Evaluation

1. Start.
2. Declare length, width, radius, rectangleArea, circleArea and totalArea.
3. Define PI = 3.14.
4. Input length and width.
5. Calculate radius = width / 2.
6. Calculate rectangleArea = length × width.
7. Calculate circleArea = $\text{PI} \times \text{radius} \times \text{radius}$.
8. Calculate totalArea = rectangleArea + circleArea.
9. Display totalArea.
10. Stop.

Evaluation: The two semicircular ends together form one complete circle. Adding its area to the rectangular area gives the total area, and the results match the test cases.

4. Flowchart and Evaluation

Flowchart logic: Start → Input length and width → Calculate radius → Calculate rectangle area → Calculate area of the two semicircles → Calculate total area → Display total area → Stop.



Evaluation: The flowchart follows the same calculation sequence as the algorithm and gives the expected total area for the prepared test cases.

5. C Program

```
#include <stdio.h>

#define PI 3.14

int main()
{
    float length, width, radius;
    float rectangleArea, circleArea, totalArea;

    printf("Enter length of rectangle: ");
    scanf("%f", &length);

    printf("Enter width of rectangle: ");
    scanf("%f", &width);

    radius = width / 2;
    rectangleArea = length * width;
    circleArea = PI * radius * radius;
    totalArea = rectangleArea + circleArea;

    printf("Total Area = %.2f sq. units\n", totalArea);

    return 0;
}
```

6. Compilation, Execution and Test Evaluation

Compilation Command:

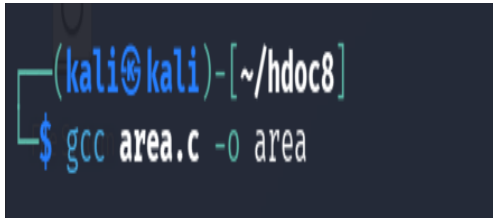
```
cc area.c -o area
```

Execution Command:

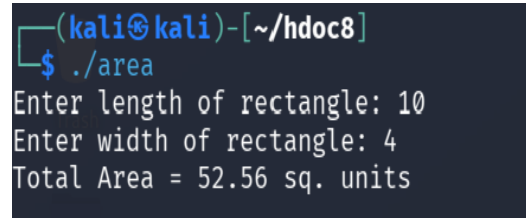
```
./area
```

The program output should be compared with the expected values in the test case table. Matching values are marked PASS.

7. Compilation and Execution

A terminal window with a dark background. The prompt is (kali@kali)-[~/hdoc8]. The command gcc area.c -o area is entered and executed.

```
(kali@kali)-[~/hdoc8]  
$ gcc area.c -o area
```

A terminal window with a dark background. The prompt is (kali@kali)-[~/hdoc8]. The command ./area is entered and executed, followed by two prompts for length and width, and the final output of the total area.

```
(kali@kali)-[~/hdoc8]  
$ ./area  
Enter length of rectangle: 10  
Enter width of rectangle: 4  
Total Area = 52.56 sq. units
```